FLEX-DBS manuscript: working notes

Working notes

Data still needed

- **Electrode construction (Section 3.1).** Each hemisphere has one electrode, a twisted pair with one stimulating wire (connected to the channel output) and one ground wire (connected to the device ground), and no separate return electrode (Sections 2.2, 3.1 and 3.2 and the Figure 1 legend). Still needed: electrode exposed surface area (or exposed length after cutting and sanding), spacing of the two hemispheric electrodes, and the connector and its attachment to the wires. The Figure 1 image still labels the electrode interface "Bilateral electrodes, common return"; relabel it (for example, "One electrode per hemisphere: stimulating wire and ground wire") to match the text. Possible figure: electrode fabrication.
- **Surgery, animals and ethics (Section 3.1).** Still needed: analgesia, the IACUC protocol number (the Ethics statement gives the approval without a number), and the implantation date that anchors "weeks after implantation". The mice in Section 3.2 and Section 6 are different animals; confirm that the coordinates and the 7–10 day recovery in Section 3.1 apply to both groups.
- **In vivo validation (Section 6).** Figure 6, its legend, the analysis methods (Section 6.2) and the results (Section 6.3) come from the mouseHat-openField export (`exports/flex_dbs_fig6`), recorded 2026-10-03 in six CCI mice (F1L, F1R, FLR, M1L, M1R, MLR). Not recorded in that project and still needed: the exact amplitude setting in each mouse (Section 6.1 says only "above 100 µA"), sex (the F/M prefix of the IDs is not documented as sex), age and strain of these mice, the CCI side, date and time since injury, the CCI surgical methods (the surgery text supplied so far covers only electrode construction and implantation, which Section 3.1 already contains; Section 6.1 mentions the CCI without a procedure), how DBS was switched on and off (the video shows a hand entering the chamber), camera model, chamber material, lighting, Bonsai version and tracking workflow, the Python version of the primary preprocessing pipeline, and experimenter blinding (none described). Firmware v0.5 is stated as in the rest of the paper, and the Section 3.2 and Section 6 mice are stated to be different animals. Electrode impedance for these six mice was not checked; reporting it would give the delivered current directly.
- **Acclimation.** Section 6.1 keeps the 5 minute acclimation as supplied, but the analysis project records it differently: its recording protocol introduces a 5 minute in-arena habituation only for recordings made after 2026-10-03, its configuration states that the animal was placed in the arena at the start of recording, and no handling event was detected near the start of any video. Confirm before posting.
- **Figure 6A timeline.** Panel A draws 5 minute boundaries (0, 5, 10, 15 min), while DBS actually started at 5.5–6.9 min and stopped at 11.7–14.4 min. The legend now calls the timeline approximate and points to Sections 6.1 and 6.2 for the actual timing; redrawing the panel is optional.
- **Consistency of the high-amplitude claim.** Section 6.1 now states the amplitude as a setting above 100 µA, with delivered current limited by compliance to about 81–101 µA at the measured loads (Figure 4B). Confirm the setting. "Supratherapeutic" was removed throughout, because delivered current was not measured in these mice; the text now refers to high amplitude settings intended to evoke a robust stimulation-associated behavioral phenotype, and the afterdischarge/freezing premise is stated as a hypothesis without a citation. The 62.7 % figure is the relative change in the group means, (75.675 –

46.504)/46.504; the mean of the per-animal relative changes is 58.8 % (SEM 22.8 %), and the absolute difference is 29.2 percentage points.

- **Firmware charge imbalance.** Section 6 used firmware v0.5 (net charge 26–35 % of the stimulating-phase charge per pulse, Section 4.1); this is listed briefly under Limitations and not discussed further.
- **Abstract and Conclusions.** The in vivo sentences supplied by R. Graham in the abstract Results and Conclusions and in Section 8 were reworded as associations rather than causal effects, because the single session without a sham control cannot separate stimulation from within-session habituation (DBS versus the final 2 min of baseline: $p = 0.066$), and only high amplitude settings were tested, so the claim of therapeutic and supratherapeutic amplitudes was dropped. The Conclusions now say "consistent with effective stimulation through implanted electrodes" and note that delivered current was not measured. The Limitations paragraph now mentions the in vivo design. Check that the wording is acceptable to all authors.

Impedance measurement

- Data come from the platinum cohort 1 (M1, M2) and platinum cohort 2 (F1, F2) sheets. Excluded: M1 output 0 (near 0 V throughout, likely a short) and both electrodes of F1 (unstable; 47–118 kΩ and 23–106 kΩ between sessions, with within-session drift at 7 weeks). Analyzed: five electrodes from M1, M2 and F2. Confirm the wording of the exclusion criteria, and consider stating a quantitative criterion.
- The StimJim bench check (2/10/20/68 kΩ) showed its voltage read-back 6–73 % below the oscilloscope, which biases the in vivo values low, while the 450 µs pulse biases them high. Consider remeasuring with an impedance meter (for example, a NanoZ at 1 kHz) or with an oscilloscope on the StimJim output, and report both.
- The 450 µs measurement pulse is longer than the DBS pulses; a measurement at the stimulation pulse width would better match the compliance calculation.
- The two-electrode configuration is justified with Koo and Weiland (2018); confirm that this is the framing the authors want.

Citations to add or verify

- Shirvalkar et al. (2025) is a medRxiv preprint (posted 2025-08-13); update if it is published. Check that it and Boccard et al. (2017) are the intended citations for cingulate DBS for refractory chronic pain (Boccard et al. report anterior cingulate DBS for neuropathic pain).
- Paulat et al. (2026) is still a Research Square preprint (posted 2026-04-27; checked 2026-10-01). Update if it is published.

Table 4

- Check whether the Pinnell et al. (2018) Figshare release (described as design files and assembly instructions) contains firmware source.
- Supplementary files were not read for Burton et al. (2021) or Shen et al. (2026), and Paulat et al. (2026)'s programmable ranges and battery models come from its supplement.

Firmware and design

- Firmware v0.5 delivers the 130 Hz setting as 142.5 Hz, and at 180 µs the stimulating phase measures ∼60 µs long and the recovery phase 22–46 µs short, leaving a 26–35 % net charge per pulse (Section 4.1). If the fixes listed in Section 7 are made before posting, remeasure and update Section 4.1, Figure 5, Table 2 and the Limitations.
- The photograph in Figure 2A shows the "v8.1" board silkscreen; decide whether to mention the hardware revision in the text.
- Supplementary S1: consider adding the BOM (`v8.1/BOM.csv`) alongside the schematic.

End matter

- Confirm author order, middle initials and affiliations.
- Funding: check R01DA0625509, which has seven digits after "DA" where NIH serial numbers have six.
- Data availability: firmware and app repositories (S2–S4) are public and linked; the S1 schematic, data and analysis code are available on request. Journals increasingly require a public repository for data.
- Before journal submission (not needed for bioRxiv): Highlights, graphical abstract, CRediT statement, generative-AI declaration, line numbers and double spacing per the Journal of Neuroscience Methods guide.

References

Boccard, S.G.J., Prangnell, S.J., Pycroft, L., Cheeran, B., Moir, L., Pereira, E.A.C., Fitzgerald, J.J., Green, A.L., Aziz, T.Z., 2017. Long-term results of deep brain stimulation of the anterior cingulate cortex for neuropathic pain. World Neurosurgery 106, 625–637. doi:10.1016/j.wneu.2017.06.173.

Burton, A., Won, S.M., Sohrabi, A.K., Stuart, T., Amirhossein, A., Kim, J.U., Park, Y., Gabros, A., Rogers, J.A., Vitale, F., Richardson, A.G., Gutruf, P., 2021. Wireless, battery-free, and fully implantable electrical neurostimulation in freely moving rodents. Microsystems & Nanoengineering 7, 62. doi:10.1038/s41378-021-00294-7.

Koo, B., Weiland, J., 2018. Experimental factors effecting stability of electrochemical impedance spectroscopy measurements, in: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 2949–2952. doi:10.1109/EMBC.2018.8512973.

Paulat, R., Koschützke, L., Haumesser, J., Maurer, L., Altschüler, J., Tang, H., Hounchonou, H., Daugardt, A., Harnack, D., Just, V., Schulz, G., Harms, C., Kühn, A.A., Wenger, N., van Riesen, C., 2026. Wireless lightweight implantable pulse generators enable chronic dual-target deep brain stimulation in rodent disease models. Research Square preprint. doi:10.21203/rs.3.rs-9019884/v1.

Pinnell, R.C., Pereira de Vasconcelos, A., Cassel, J.C., Hofmann, U.G., 2018. A miniaturized, programmable deep-brain stimulator for group-housing and water maze use. Frontiers in Neuroscience 12, 231. doi:10.3389/fnins.2018.00231.

Shen, Y., Zhao, D., Lv, Y., Liu, Y., Chen, Y., Zhang, Y., Chen, S., 2026. A wearable deep brain stimulation system for behavioral studies in rodents. Frontiers in Neuroscience 19, 1707636. doi:10.3389/fnins.2025.1707636.

Shirvalkar, P., Leriche, R., Saal, J., Cagle, J., Prosky, J., Joseph, I., Shaughnessy, A., Schmitgen, A., Lin, J., Motzkin, J., Dawes, H., Racine, C., Seritan, A., Sellers, K.K., deHemptinne, C., Starr, P.A., Chang, E.F., 2025. Personalized, closed-loop deep brain stimulation for chronic pain. medRxiv preprint. doi:10.1101/2025.08.11.25333010.